

SQL Interview Questions – Day 2

5 more SQL interview questions with answers and examples.

1. What is a UNIQUE constraint in SQL?

Answer: A UNIQUE constraint ensures that values in a column are different. A table can have multiple UNIQUE constraints.

```
CREATE TABLE employees (  
  employee_id INT PRIMARY KEY,  
  email VARCHAR(100) UNIQUE  
);
```

2. What is the difference between DELETE, TRUNCATE, and DROP?

Answer: DELETE removes selected rows and can use WHERE. TRUNCATE removes all rows but keeps the table structure. DROP removes the table and its structure.

```
DELETE FROM employees WHERE employee_id = 10;  
TRUNCATE TABLE employees;  
DROP TABLE employees;
```

3. What is a NULL value in SQL?

Answer: NULL represents a missing, unknown, or not-applicable value. It is not the same as 0 or an empty string. Use IS NULL or IS NOT NULL to check it.

```
SELECT *  
FROM employees  
WHERE manager_id IS NULL;
```

4. What is an aggregate function in SQL?

Answer: Aggregate functions calculate values from multiple rows. Common examples are COUNT(), SUM(), AVG(), MIN(), and MAX().

```
SELECT COUNT(*) AS total_employees,  
       AVG(salary) AS average_salary,  
       MAX(salary) AS highest_salary  
FROM employees;
```

5. What is a Foreign Key?

Answer: A foreign key creates a relationship between two tables by referencing a primary key or suitable unique key in another table. It helps maintain referential integrity.

```
CREATE TABLE departments (  
  department_id INT PRIMARY KEY,  
  department_name VARCHAR(50)  
);  
  
CREATE TABLE employees (  
  employee_id INT PRIMARY KEY,  
  department_id INT,  
  FOREIGN KEY (department_id)  
  REFERENCES departments(department_id)  
);
```